

Spivak Ch 10 Q26-27

a). Suppose that  $f(x) = xg(x)$ ,  $g$  is cts. at 0

Show that  $f$  is differentiable at 0. What is  $f'(0)$ ?

b). Suppose that  $f$  is diff. at 0 &  $f'(0) = 0$

Show that  $f(x) = xg(x)$  for some  $g(x)$  cts. at 0

Pf a). By definition, if  $f(x)$  exists, then

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0} \frac{f(h) - (0 \cdot g(0))}{h} \\ &= \lim_{h \rightarrow 0} \frac{h \cdot g(h)}{h} \\ &= \lim_{h \rightarrow 0} g(h) \\ &= g(0)\end{aligned}$$

b/c.  $g$  is cts. the limit exists &  $f'(0) = g(0)$ .

b). Claim that we need  $g(x) = \begin{cases} \frac{f(x)}{x}, & x \neq 0 \\ f'(0), & x = 0 \end{cases}$

For continuity, want this to be

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{f(0+x) - f(0)}{x} = f'(0)$$

To show the continuity of  $g$ , using  $\epsilon$ - $\delta$ :

let  $\epsilon > 0$ , want  $\delta$  s.t. for  $|x| < \delta$ ,  $|g(x) - f'(0)| < \epsilon$

• If  $x = 0$ , then  $|g(0) - f'(0)| = 0 < \epsilon$ .

• If  $x \neq 0$ , then  $|g(x) - f'(0)| = \left| \frac{f(x)}{x} - f'(0) \right|$

But since  $f'(c) = \lim_{h \rightarrow 0} \frac{f(h) - f(c)}{h}$ ,  $\exists \delta$  s.t.  $|\frac{f(h) - f(c)}{h} - f'(c)| < \epsilon \quad \forall |h| < \delta$   
 So take this  $\delta$



Ex. Let  $f(x) = x^3 - x^2 - 8x + 1$  on  $[-2, 2]$

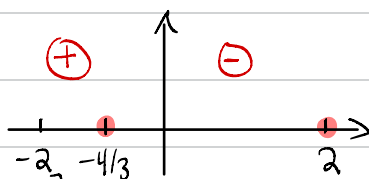
Sketch the graph of  $f(x)$

① Critical pts.  $f'(x) = 3x^2 - 2x - 8 = (3x+4)(x-2)$

$$x = \underbrace{-2, -4/3}_{f'(x)=0}, \underbrace{-2}_{\text{endpoint}}$$

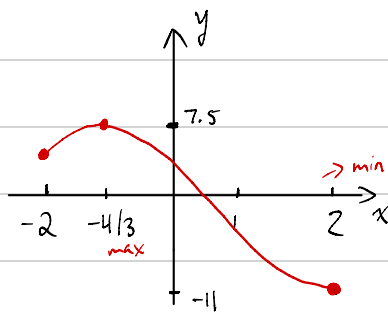
② First Derivative Test: If  $f'(x) \geq 0$  on  $[a, b]$ , then it is increasing.  
 (decreasing)  $(f'(x) \leq 0)$

$f'(x)$  looks like



In the interval  $[-4/3, 2]$

1st derivative is either  $> 0$   
 $< 0$



If  $x, y \in (-4/3, 2)$  s.t.  $f'(x) > 0 > f'(y)$ . then by IVT,  $\exists z \in (-4/3, 2)$  s.t.  $f'(z) = 0$ , contradiction

$$f'(-2) = 8 \quad f'(0) = -8$$

Since  $f'(x)$  is cts. test a value in each of  $(-2, -4/3)$  &  $(-4/3, 2)$

Ex Sketch the graph of  $f(x) = \frac{x^2 - 2x + 2}{x-1}$  on  $\mathbb{R}$

Soln ① Find the critical pts

$$f'(x) = \frac{(x-1)(2x-2) - (x^2 - 2x + 2)}{(x-1)^2}$$
$$= \frac{x^2 - 2x}{(x-1)^2}$$

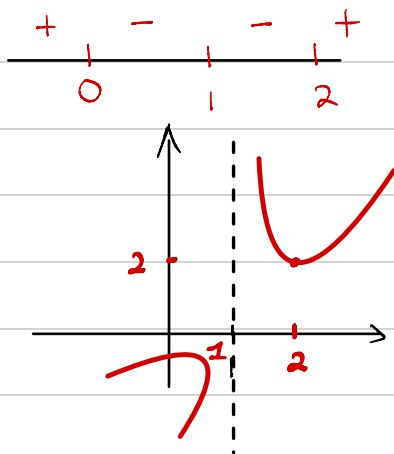
Critical pts:  $x = 0, 1, 2$

Thm 2<sup>nd</sup> Derivative Test

Let  $a$  be a critical pt for  $f$ :

If  $f''(a) > 0 \Rightarrow f$  has a local min at  $a$

If  $f''(a) < 0 \Rightarrow f$  has a local max at  $a$



Pf WOLOG let  $f''(a) > 0$

By definition,  $f''(a) = \lim_{h \rightarrow 0} \frac{f'(a+h) - f'(a)}{h} = \lim_{h \rightarrow 0} \frac{f'(a+h)}{h} > 0$

As  $h \rightarrow 0^+$ ,  $f'(a+h) > 0$

As  $h \rightarrow 0^-$ ,  $f'(a+h) < 0$

Since  $f''$  exists,  $f'$  is cts at  $a$

So  $a$  is a local minimum.



## Thm Converse of 2<sup>nd</sup> Derivative Test

If  $f''(a)$  exist &  $f$  has a local min at  $a$ , then  $f''(a) \geq 0$

max

$$f''(a) \leq 0$$

Note We have inconclusive tests as well

Ex  $f(x) = x^4$

$$f'(x) = 4x^3$$

$$f''(x) = 12x^2$$

$$f''(0) = 0$$



pf Suppose that  $a$  is a local min of  $f$

Suppose for a contradiction  $f''(a) \neq 0 \Rightarrow f''(a) < 0$

So  $a$  is a local max of  $f$

$\Rightarrow f$  is locally constant

i.e.  $\exists$  open interval  $I$  s.t.  $\forall x \in I$ ,  $f(x)$  constant.

$I$  = intersection of two interval where  $I_1$  is a local min in  $I_1$

$I_2$  is a local max in  $I_2$

Thm  $f$  cts. at  $a$ ,  $f'(x)$  exists in some neighborhood  $I$  of  $a$ , except maybe at  $a$ .

Suppose  $\lim_{x \rightarrow a} f'(x)$  exists. Then  $f'(a)$  also exists &  $f'(a) = \lim_{x \rightarrow a} f'(x)$

$\Rightarrow f'(a)$  never has isolated discontinuity

But it can jump if  $f(x) = |x|$ .